iv. 4 core cable
v. 3 ½ core cable

1.11.4 Three phase cable

Commonly, underground cables are used to transmit three phase power supply. For three phase service, the following cables are used.

a. Belted cable up to 11 KV
b. Screened cable 22KV to 66 KV
c. Pressure cable above 66 KV

a. Belted cable up to 11 KV

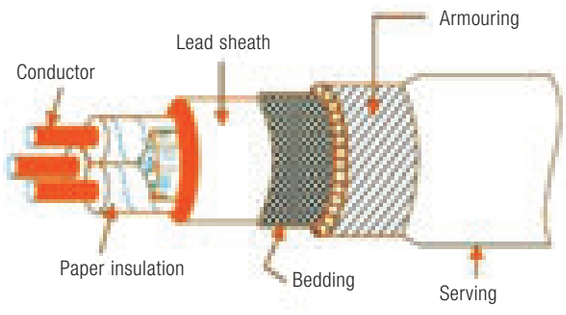


Fig. 1.17 Belted cable

Figure 1.17 shows the cross-section view of the belted cable. Each conductor is insulated by separate layers of impregnated paper. Thus, the insulated 3 conductors are set up as a part in the same set and then wrapped in an impregnated paper tape called a paper belt. Between insulated containers, it is filled with insulating material like jute or fiber. A lead sheath is laid on the paper bar to protect from mechanical damage and moisture. In this one or more layers are shielded and protected. These types of electric cables are used to transmit voltage up to 11 KV and sometimes voltage can be extended upto 22 KV.

b. Screened cable

It is of two types

i. H – type cable (Hochstadter type)
ii. S.L. type cable (Separate lead type)

i. H Type cables 22 KV to 66 KV

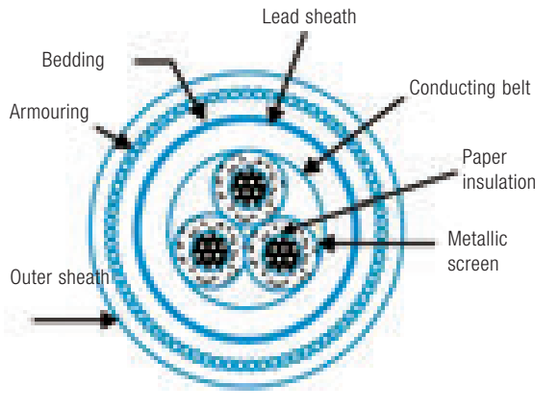


Fig. 1.18 H type cable

Figure 1.18 shows the H type screened cable. Each conductor is insulated by separate layers of impregnated paper. Then it will be wrapped with metal insulation separately. The metal shield is made of thin aluminum and made to touch each other. These three containers are wrapped in copper fabric tape. There will be a lead sheath casing on it. There is one or more armors on it. These types of cables are used to transmit low and medium voltages. It is used to transmit 33 KV and sometimes to carry up to 66 KV.

ii. S.L. type cable (22KV to 66KV)

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